
Gaël-Pacôme Nguimeya Tematio

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PROFILE

I am a researcher in computational condensed matter and materials physics with expertise in Density Functional Theory (DFT), electronic structure calculations, and topological materials. My broader research interests lie at the interface of quantum physics and quantum computing, particularly quantum algorithms, quantum simulation, and quantum machine learning.

TECHNICAL SKILLS

DFT & Tight-Binding Modeling: Proficient in Quantum Espresso, Wannier90, WannierTools, and Quantum Lattice.

Crystallography & Visualization: Skilled in Xcrysden, FermiSurfer, Vesta, and Avogadro for molecular and crystal structure analysis.

Quantum Computing: Experienced with industry-standard frameworks including Qiskit, PennyLane, Classiq, and Cirq.

Programming & Version Control: Advanced knowledge of Python and Bash scripting; proficient in Git, GitHub, Google Apps Script, and LaTeX (including Beamer).

Operating Systems: Full competency in both Linux and Windows environments.

EDUCATION & TRAINING

Ph.D., Computational Condensed Matter and Materials Physics, University of Cape Town, Sept 2025
South Africa

Research Master, Condensed Matter Physics, University of Dschang, Cameroon Sept 2015

Structured Master, Mathematical Sciences, the African Institute for Mathematical Sciences (AIMS) Cameroon Jun 2015

Bachelor of Science, Condensed Matter Physics, University of Dschang, Cameroon Sept 2012

LIST OF PUBLICATIONS

- [3] Gaël-Pacôme Nguimeya Tematio, Trisha Salagaram, Aniekan Magnus Ukpog, Houssam Eddine Hailouf, and Kingsley Onyebuchi Obodo. Strain and electric field control of topological phases in $SrSi_2$ and $CoSi$: A first-principles and model Hamiltonian study. *Mater. Chem. Phys.* 132574 (2026). <https://doi.org/10.1016/j.matchemphys.2026.132574>
- [2] Gaël-Pacôme Nguimeya Tematio, Trisha Salagaram, Aniekan Magnus Ukpog, Houssam Eddine Hailouf, and Kingsley Onyebuchi Obodo. Field-driven electronic and topological modulation in the Weyl semimetal NbP. *J. Phys.: Condens. Matter* **38**, 115501 (2026). <https://doi.org/10.1088/1361-648X/ae50a2>
- [1] Kenfack SC, Nguimeya GP, Talla PK, Fotue AJ, Fobasso MFC, Tiotsop M, Djomou JRD, Ekosso CM and Fai LC. Decoherence of driven coupled harmonic oscillator. *J. Nanosci. Curr. Res.* **1**, 1000104 (2016). <https://doi.org/10.4172/2572-0813.1000104>

TEACHING

Teaching assistant (Head Tutor), African Institute for Mathematical Sciences (AIMS), South Africa	<i>Sept 2022 - Oct 2023</i>
Teaching assistant (Tutor), African Institute for Mathematical Sciences (AIMS), South Africa	<i>Feb 2022 - Jun 2022</i>
Teaching assistant, University of Cape Town (UCT), South Africa	<i>Feb 2019 - Nov 2019</i>
Teaching assistant, Higher Teacher Training College (HTTC) Bambili, Cameroon.	<i>Feb 2017 - Mar 2017</i>
• Quantum Mechanics (Undergraduate course). • Advanced Quantum Mechanics (Graduate course).	
Part-time High School Physics Teacher, Government Bilingual High School (GBHS) Mbouda, Cameroon	<i>Sept 2016 - Jun 2017</i>
Teaching assistant, University of Dschang, Cameroon.	<i>May 2016 - Jun 2016</i>
• Quantum mechanics (Graduate course).	
Teaching assistant, Higher Teacher Training College (HTTC) Bambili, Cameroon.	<i>Feb 2016 - Mar 2016</i>
• Quantum mechanics (Graduate course).	
Teaching assistant, University of Dschang, Cameroon.	<i>Nov 2015 - Jan 2016</i>
• Electromagnetism (Undergraduate course). • Electrostatics (Undergraduate course). • General Physics (Undergraduate course).	
Teaching assistant, University of Dschang, Cameroon.	<i>May 2014 - Jul 2014</i>
• Introduction to Quantum Physics (Undergraduate course).	

CERTIFICATIONS

Badge Womanium 2025 Quantum Program issued by The Washington Institute for STEM, Entrepreneurship and Research.	<i>Aug 2025</i>
Quantum Algorithms to Solve Partial Differential Equations issued by The Washington Institute for STEM, Entrepreneurship and Research.	<i>Aug 2025</i>
Quantum Algorithms for Nonlinear Problems issued by The Washington Institute for STEM, Entrepreneurship and Research.	<i>Aug 2025</i>
PennyLane Trotterization Challenge at the Womanium & Wiser Quantum Program 2025 issued by PennyLane.	<i>Aug 2025</i>
Qiskit Global Summer School 2025 - Quantum Excellence Summer School by IBM Quantum.	<i>Aug 2025</i>
PennyLane Linear Combination of Unitaries (LCU) Challenge at the Womanium & Wiser Quantum Program 2025 issued by PennyLane.	<i>Jul 2025</i>
Quantathon (Hackathon at AIMS Ghana) by AIMS Ghana QTedu Global Event: a week-long celebration of innovation, collaboration, and quantum education for the SDGs; awarded second place to my team.	<i>Jul 2025</i>
Full list on LinkedIn: https://www.linkedin.com/in/pacomenguimeya .	

LEADERSHIP ROLES

Class Representative: Class representative for the MSc in Condensed Matter Physics at the University of Dschang, Cameroon.

Oct 2013 - Sept 2015

LANGUAGES

French: Mother tongue

English: Fluent

REFERENCES

Available upon request.